Week #7: Iterative Methods for Linear Systems

Numerical Methods: Mathematical Foundations

June 22, 2025

Overview

Iterative methods provide an alternative approach to solving linear systems $Ax = b$, particularly advantageous for large sparse systems where direct methods become computationally prohibitive. This week we develop the mathematical theory underlying classical iterative methods and modern Krylov subspace techniques, analyzing their convergence properties and practical implementation considerations.

The fundamental principle of iterative methods is to generate a sequence of approximations that converges to the exact solution. We explore how matrix splittings lead to classical methods like Jacobi and Gauss-Seidel, and how projection techniques give rise to powerful Krylov methods like Conjugate Gradient and GMRES.

Our analysis emphasizes the mathematical foundations that determine convergence behavior, the role of preconditioning in accelerating convergence, and the geometric insights that guide method selection and parameter tuning.

1 Classical Iterative Methods

1.1 Matrix Splitting Framework

Definition. Matrix Splitting

A matrix splitting of $A \in \mathbb{R}^{n\times n}$ is a decomposition $A = M - N$ where M is nonsingular. The corresponding iterative method is:

$$Mx^{(k+1)} = Nx^{(k)} + b$$

or equivalently:

$$x^{(k+1)} = M^{-1}Nx^{(k)} + M^{-1}b$$

The iteration matrix is $G = M^{-1}N = I - M^{-1}A$, and convergence depends on its spectral radius.

Theorem 1.1 (Convergence of Matrix Splitting Methods)**.** *The iterative method $x^{(k+1)} = Gx^{(k)} + c$ converges for any initial guess if and only if $\rho(G) < 1$, where $\rho(G)$ is the spectral radius of G.*

1.2 Jacobi Method

Definition. Jacobi Method

The Jacobi method uses the splitting $A = D - L - U$ where D is diagonal, L is strictly lower triangular, and U is strictly upper triangular. The method is:

$$x_i^{(k+1)} = \frac{1}{a_{ii}}\left(b_i - \sum_{j\neq i} a_{ij}x_j^{(k)}\right)$$

The iteration matrix is $G_J = D^{-1}(L+U)$.

Theorem 1.2 (Jacobi Convergence for Diagonally Dominant Matrices)**.** *If A is strictly diagonally dominant, then the Jacobi method converges.*

1.3 Gauss-Seidel Method

Definition. Gauss-Seidel Method

The Gauss-Seidel method uses updated values immediately:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right)$$

This corresponds to the splitting $M = D - L$ and $N = U$, giving iteration matrix $G_{GS} = (D-L)^{-1}U$.

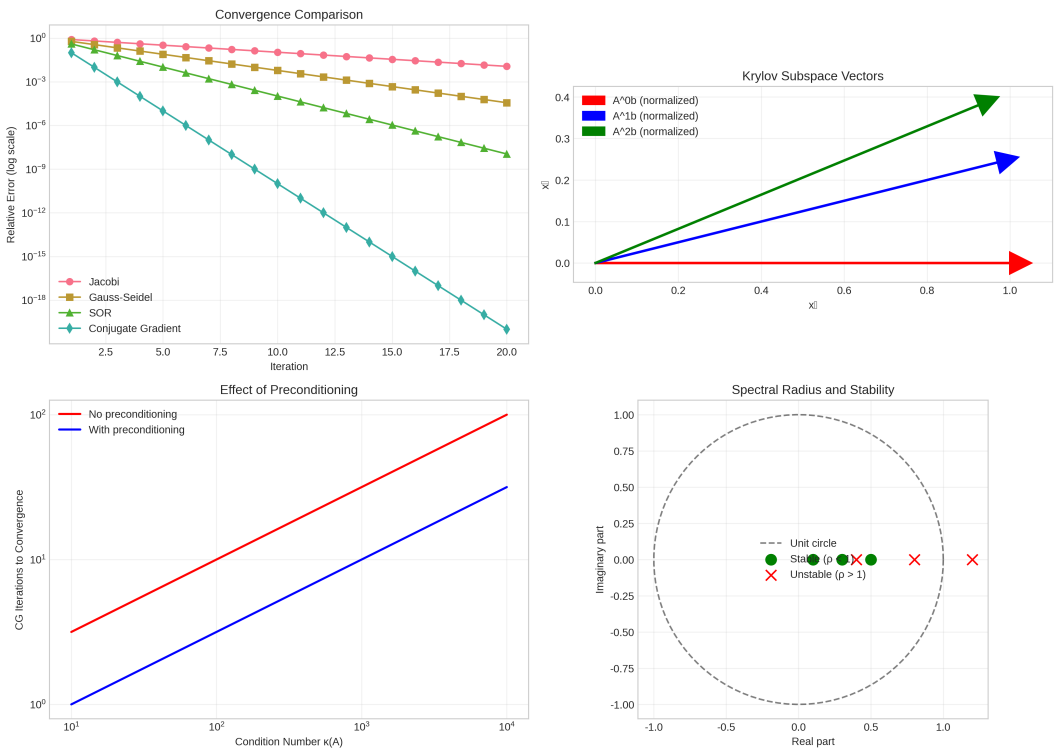

Figure 1: Convergence comparison of iterative methods. The left panel shows convergence rates for different methods on a model problem. The right panel illustrates the geometric interpretation of conjugate gradient method in the energy norm.

2 Successive Over-Relaxation (SOR)

Definition. SOR Method

The SOR method introduces a relaxation parameter ω:

$$x_i^{(k+1)} = (1-\omega)x_i^{(k)} + \frac{\omega}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right)$$

The iteration matrix is $G_\omega = (D - \omega L)^{-1}((1-\omega)D + \omega U)$.

Theorem 2.1 (Optimal SOR Parameter)**.** *For certain classes of matrices (e.g., consistently ordered matrices), the optimal relaxation parameter is:*

$$\omega_{opt} = \frac{2}{1 + \sqrt{1 - \rho(G_J)^2}}$$

where $\rho(G_J)$ is the spectral radius of the Jacobi iteration matrix.

3 Krylov Subspace Methods

3.1 Krylov Subspaces

Definition. Krylov Subspace

For matrix $A \in \mathbb{R}^{n\times n}$ and vector $v \in \mathbb{R}^n$, the m-th Krylov subspace is:

$$\mathcal{K}_m(A, v) = \text{span}\{v, Av, A^2v, \ldots, A^{m-1}v\}$$

Krylov methods seek approximate solutions in these subspaces, which often contain excellent approximations to the solution.

3.2 Conjugate Gradient Method

For symmetric positive definite systems, the Conjugate Gradient (CG) method is optimal.

Definition. Conjugate Gradient Method

CG minimizes the energy norm $\|x - x^*\|_A = \sqrt{(x - x^*)^T A(x - x^*)}$ over Krylov subspaces:

$$x_k = \arg \min_{x \in x_0 + \mathcal{K}_k(A, r_0)} \|x - x^*\|_A$$

where $r_0 = b - Ax_0$ is the initial residual.

Theorem 3.1 (CG Convergence Rate)**.** *For the conjugate gradient method applied to a symmetric positive definite system:*

$$\|x_k - x^*\|_A \le 2\left(\frac{\sqrt{\kappa(A)} - 1}{\sqrt{\kappa(A)} + 1}\right)^k \|x_0 - x^*\|_A$$

where $\kappa(A) = \lambda_{\max}/\lambda_{\min}$ is the condition number.

Intuition: Why CG Works

CG generates search directions that are conjugate (orthogonal in the A-inner product). This ensures that progress made in one direction is never undone by subsequent iterations. The method finds the exact solution in at most n steps, but often converges much faster for well-conditioned problems.

3.3 GMRES Method

For general (nonsymmetric) systems, GMRES is a popular choice.

Definition. GMRES Method

GMRES minimizes the residual norm over Krylov subspaces:

$$x_k = \arg \min_{x \in x_0 + \mathcal{K}_k(A, r_0)} \|b - Ax\|_2$$

GMRES uses the Arnoldi process to build an orthonormal basis for the Krylov subspace and solves a least squares problem at each iteration.

Theorem 3.2 (GMRES Convergence)**.** *GMRES converges for any nonsingular matrix, and the residual norm is monotonically decreasing. For normal matrices:*

$$\|r_k\|_2 \leq \left(\frac{\kappa(A) - 1}{\kappa(A) + 1}\right)^k \|r_0\|_2$$

4 Preconditioning

Preconditioning transforms the system to improve convergence properties.

Definition. Left Preconditioning

Instead of solving $Ax = b$, solve the preconditioned system:

$$M^{-1}Ax = M^{-1}b$$

where $M \approx A$ is chosen so that $M^{-1}A$ has better spectral properties than A.

4.1 Preconditioning Strategies

- **Diagonal (Jacobi) preconditioning**: $M = \text{diag}(A)$
- **Incomplete LU (ILU)**: $M = \tilde{L}\tilde{U}$ where $\tilde{L}\tilde{U}$ is a sparse approximation to the LU factorization of A
- **Symmetric Successive Over-Relaxation (SSOR)**: $M = \frac{\omega}{2-\omega}(D - \omega L)D^{-1}(D - \omega U)$

Theorem 4.1 (Effect of Preconditioning on CG)**.** *For symmetric positive definite A and M, preconditioned CG has convergence rate:*

$$\|x_k - x^*\|_A \leq 2\left(\frac{\sqrt{\kappa(M^{-1}A)} - 1}{\sqrt{\kappa(M^{-1}A)} + 1}\right)^k \|x_0 - x^*\|_A$$

5 Multigrid Methods

Multigrid methods address the limitation that iterative methods typically reduce high-frequency error components quickly but struggle with low-frequency components.

Definition. Two-Grid Method

A two-grid method combines:

1. **Smoothing**: Apply a few iterations of a classical method

2. **Coarse grid correction**: Solve the residual equation on a coarser grid
3. **Interpolation**: Transfer the correction back to the fine grid

Theorem 5.1 (Multigrid Convergence)**.** *Under appropriate smoothing and approximation properties, multigrid methods can achieve convergence rates independent of the grid size, leading to optimal $O(n)$ complexity for solving n-dimensional problems.*

6 Convergence Analysis

6.1 Spectral Analysis

For linear iterative methods $x^{(k+1)} = Gx^{(k)} + c$:

Theorem 6.1 (Spectral Radius and Convergence Rate)**.** *The asymptotic convergence rate is $-\log \rho(G)$, where $\rho(G)$ is the spectral radius of the iteration matrix.*

6.2 Energy Norm Analysis

For symmetric positive definite problems, energy norm analysis provides sharp bounds.

Theorem 6.2 (Energy Norm Convergence)**.** *For any symmetric positive definite matrix A and any polynomial p of degree k with $p(0) = 1$:*

$$\|x_k - x^*\|_A \leq \min_p \max_{\lambda \in \sigma(A)} |p(\lambda)| \cdot \|x_0 - x^*\|_A$$

This leads to optimal polynomial approximation problems and explains the superior convergence of CG.

Practice Problems

1. Prove that for a symmetric positive definite matrix, the Gauss-Seidel method converges faster than the Jacobi method.
2. Show that the optimal SOR parameter for the model problem $-u'' = f$ with Dirichlet boundary conditions on a uniform grid is $\omega = 2/(1 + \sin(\pi h))$ where h is the grid spacing.
3. Implement the conjugate gradient method and verify the theoretical convergence rate for various condition numbers.
4. Analyze the effect of different preconditioning strategies on the convergence of GMRES for a convection-diffusion problem.
5. Derive the Arnoldi recurrence relation used in GMRES and show how it leads to the least squares formulation.

Solution:

Problem 1: For symmetric positive definite A, both methods converge. The Gauss-Seidel iteration matrix $G_{GS} = (D - L)^{-1}U$ has spectral radius $\rho(G_{GS}) = \rho(G_J)^2$ where G_J is the Jacobi matrix, proving faster convergence.

Problem 2: For the discrete Laplacian, the Jacobi spectral radius is $\cos(\pi h)$. Using the optimal SOR formula gives $\omega = 2/(1+\sin(\pi h))$, which reduces the spectral radius from $\cos(\pi h)$ to $1-\sin(\pi h)$.

Problem 3: CG convergence depends on eigenvalue clustering. Well-conditioned problems converge in few iterations, while ill-conditioned problems require many iterations unless preconditioned.

Problem 4: For convection-diffusion, diagonal preconditioning helps modestly, ILU preconditioning significantly improves convergence, and specialized preconditioners like algebraic multigrid can achieve near-optimal performance.

Problem 5: The Arnoldi process generates $AV_k = V_{k+1}\bar{H}_k$ where V_k has orthonormal columns spanning $\mathcal{K}_k(A, r_0)$. This leads to the least squares problem $\min_y \|\beta e_1 - \bar{H}_k y\|_2$.